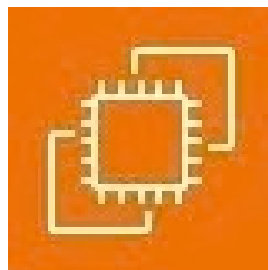




AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Capstone Projects

Trainer: Aravindraaj.G- Nminds Academy
AWS Academy Certified Educator

1.Create an EC2 Windows Instance in AWS



SHABHARIVAASUN K R
CSE-Artificial Intelligence & Machine Learning,
Sri Eshwar College of Engineering,
Coimbatore.



Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Windows Instances:

Web and Application Hosting: Run IIS web servers or other Windows-based web applications in the cloud.

Database Servers: Host SQL Server databases with EC2 Windows instances, using RDS or EC2 for more control.

Remote Desktop Applications: Set up RDS (Remote Desktop Services) for remote access to Windows desktops in the cloud.

Development and Testing: Create development environments with Windows OS to test .NET or other Windows-based applications.

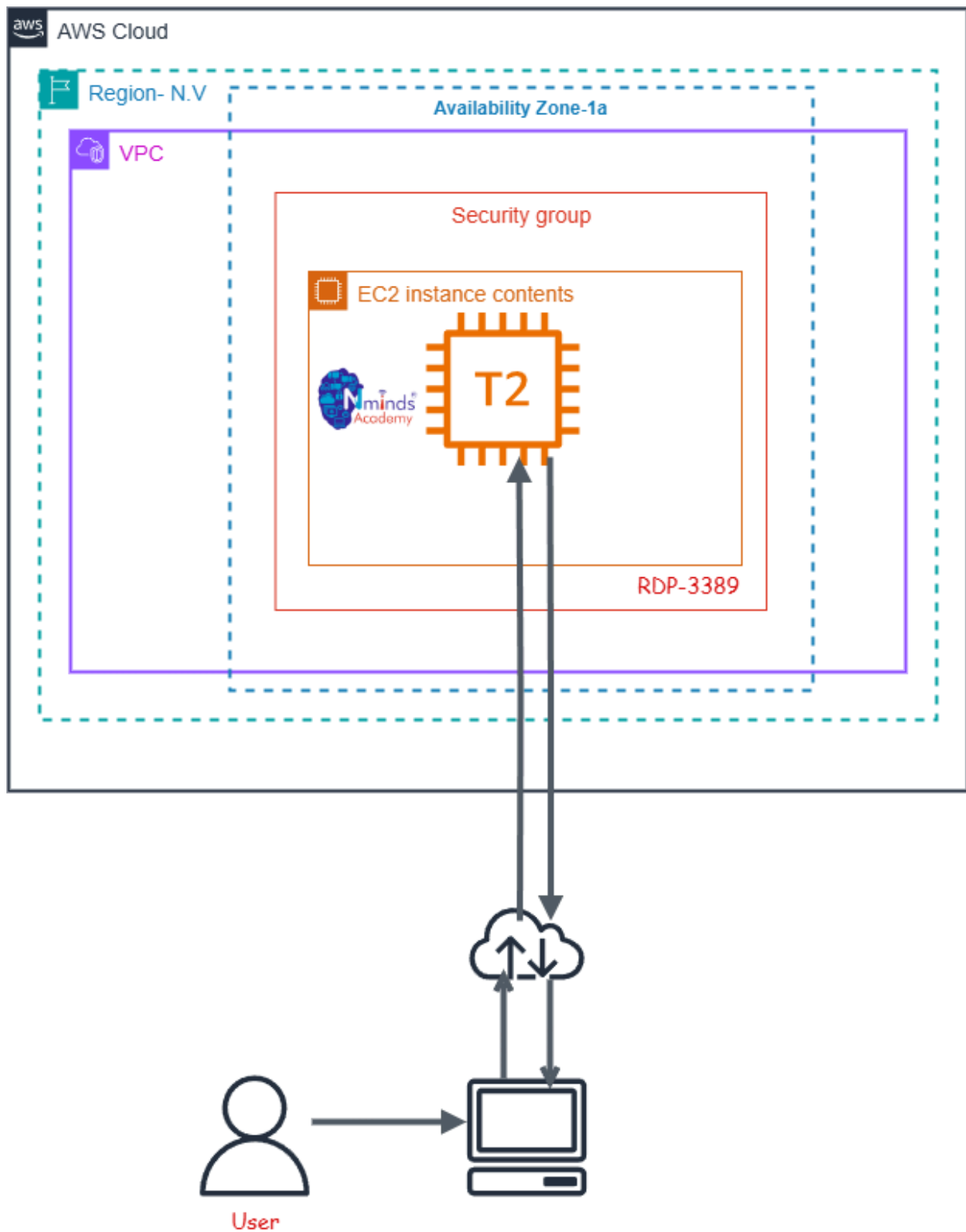
Used Services in AWS:



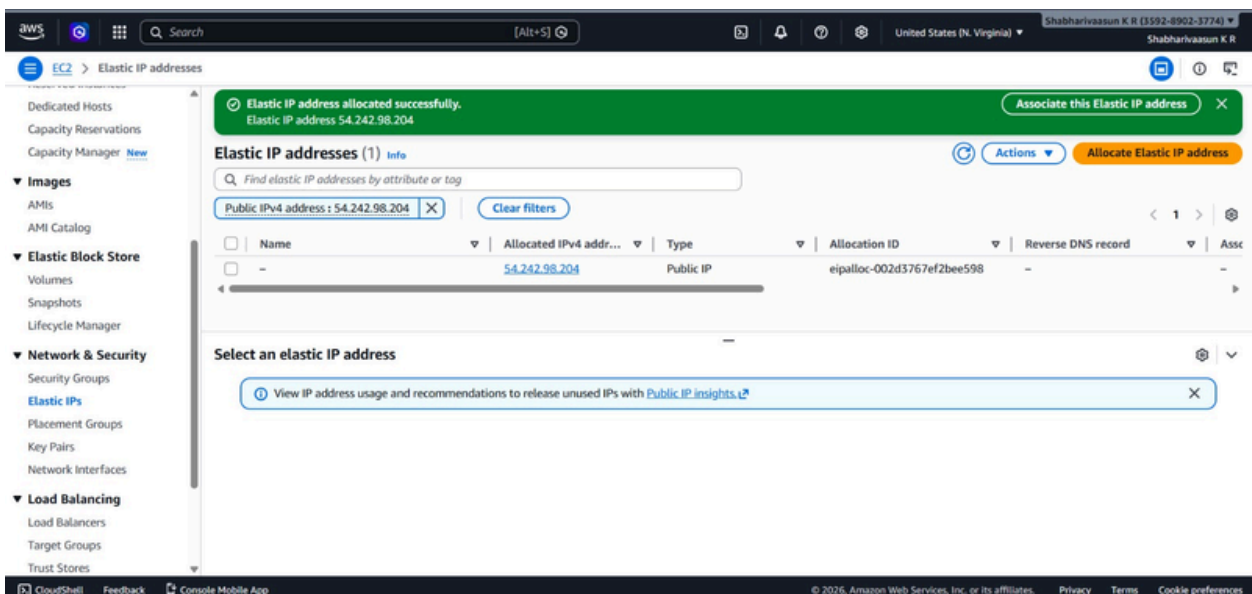
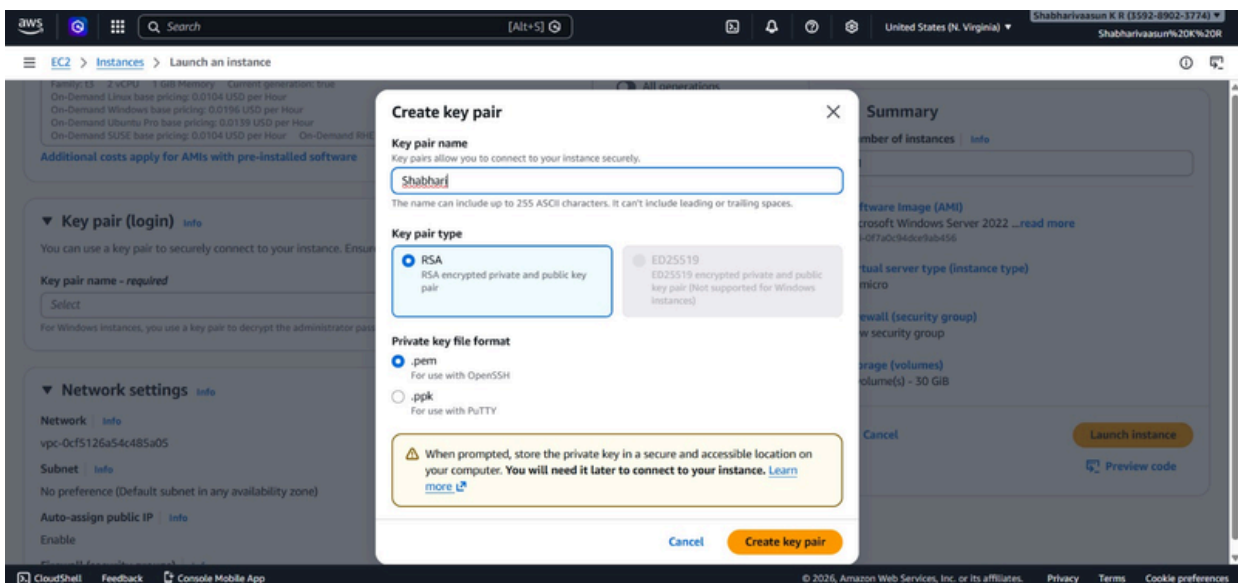
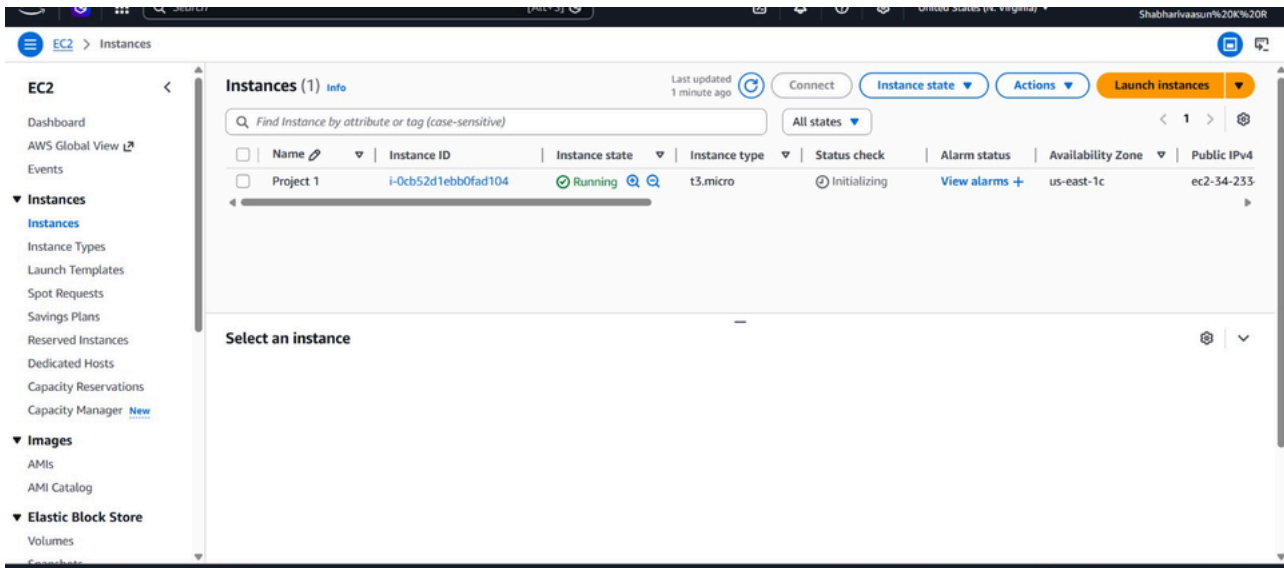
Amazon EC2



Topology



Execution Tasks:



Execution Tasks:

Success
Successfully initiated launch of Instance (i-0cb52d1ebb0fad104)

► **Launch log**

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots.
[Create EBS snapshot policy](#)

[Manage detailed monitoring](#) [Create Load Balancer](#) [Create AWS budget](#) [Manage CloudWatch alarms](#)

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Trainer: Aravindraaj.G- Nminds Academy
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2.Configure IIS Webserver on Windows Server in AWS



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Objective

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

IIS (Internet Information Services) is a web server software created by Microsoft for hosting websites and web applications on Windows Server operating systems. It provides a platform for building, deploying, and managing web-based applications. IIS supports multiple protocols, including HTTP, HTTPS, FTP, and more, and is tightly integrated with the Windows OS.

Common Use Cases of IIS:

1. **Hosting Websites:** IIS is often used by businesses and developers to host websites and web applications, particularly those built with ASP.NET.
2. **Web Applications:** For applications that require robust back-end support, IIS provides features like application pooling and security for web services (e.g., SOAP or RESTful APIs).
3. **Secure File Sharing:** IIS can be configured as an FTP server to securely share files between different locations.
4. **Application Hosting in Enterprises:** Many enterprise applications built on Microsoft technologies rely on IIS for deployment, such as SharePoint, Exchange Server, or Dynamics CRM.
5. **API Hosting:** IIS is commonly used to host APIs (e.g., RESTful services) built with .NET or other technologies.

Used Services in AWS

Used Services in Windows Server

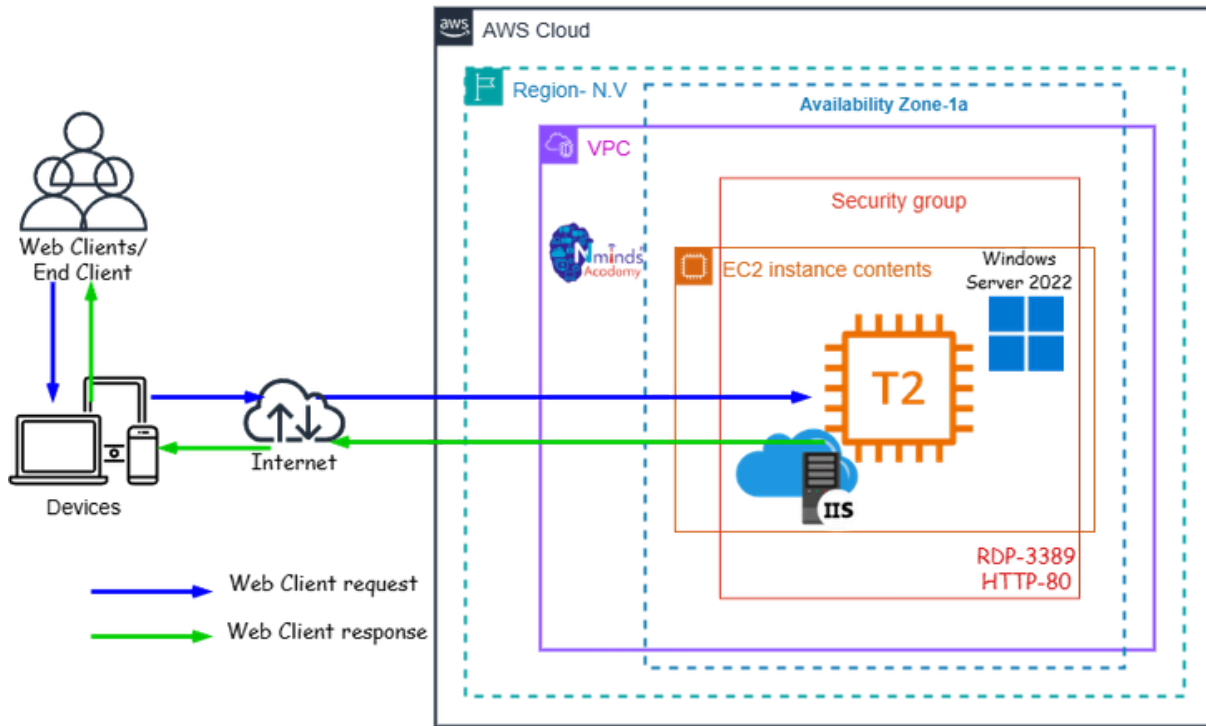


IIS- Internet Information Services

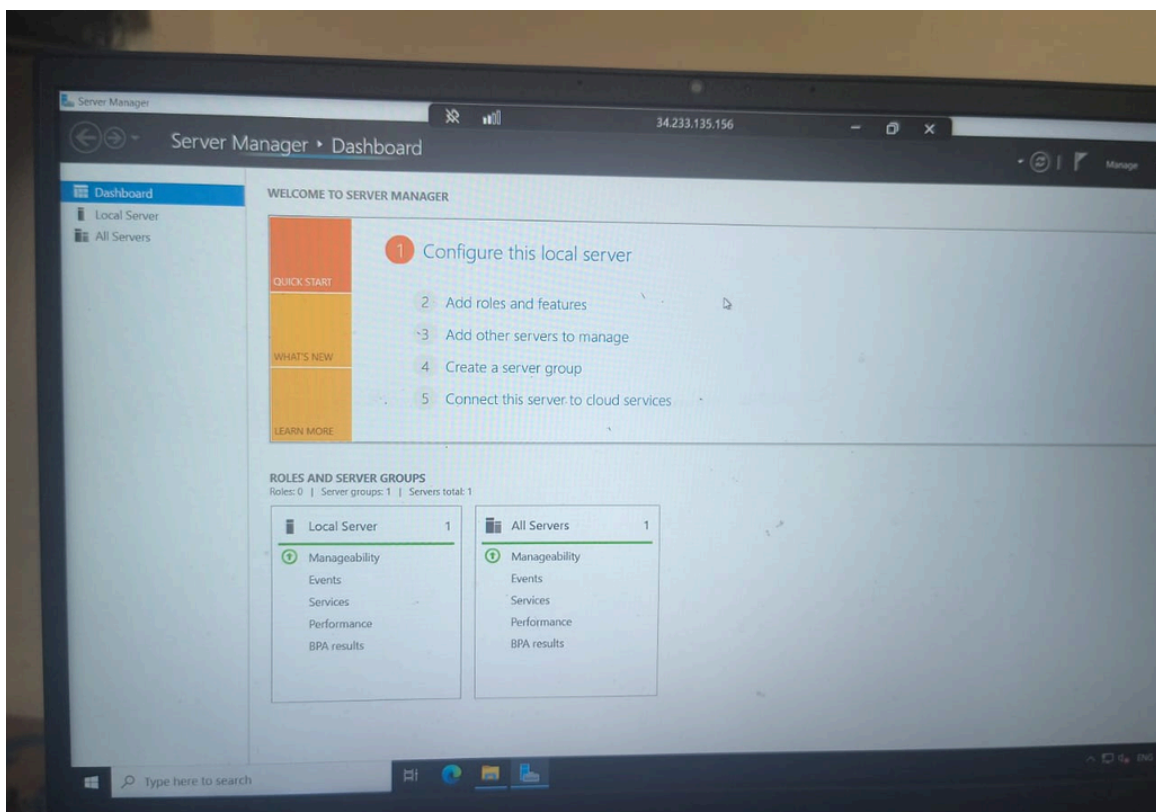
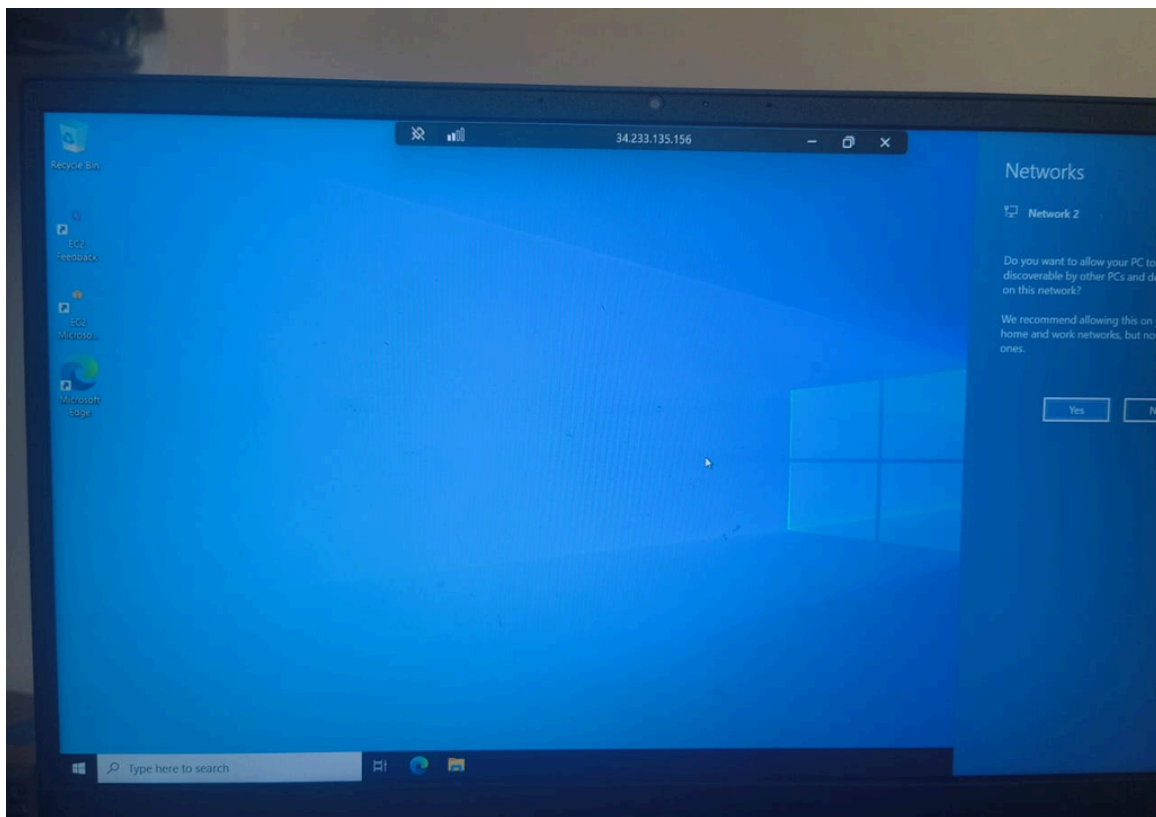


Topology

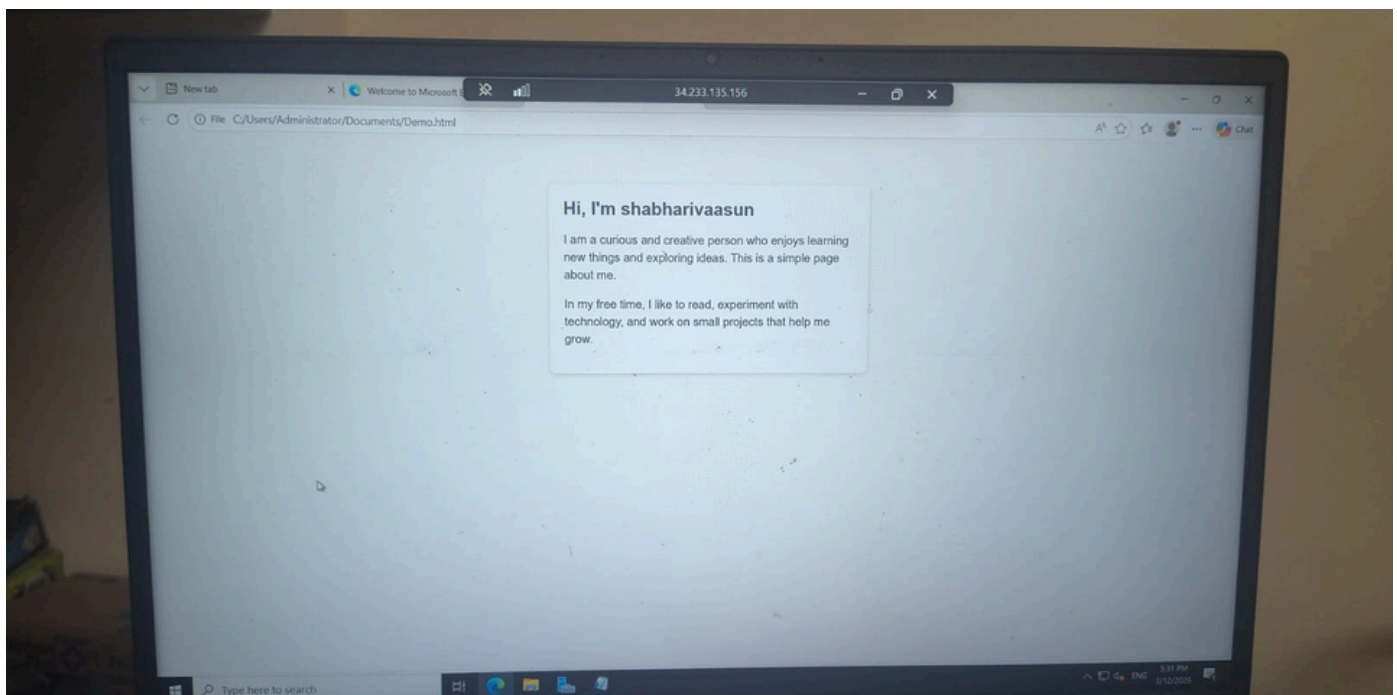
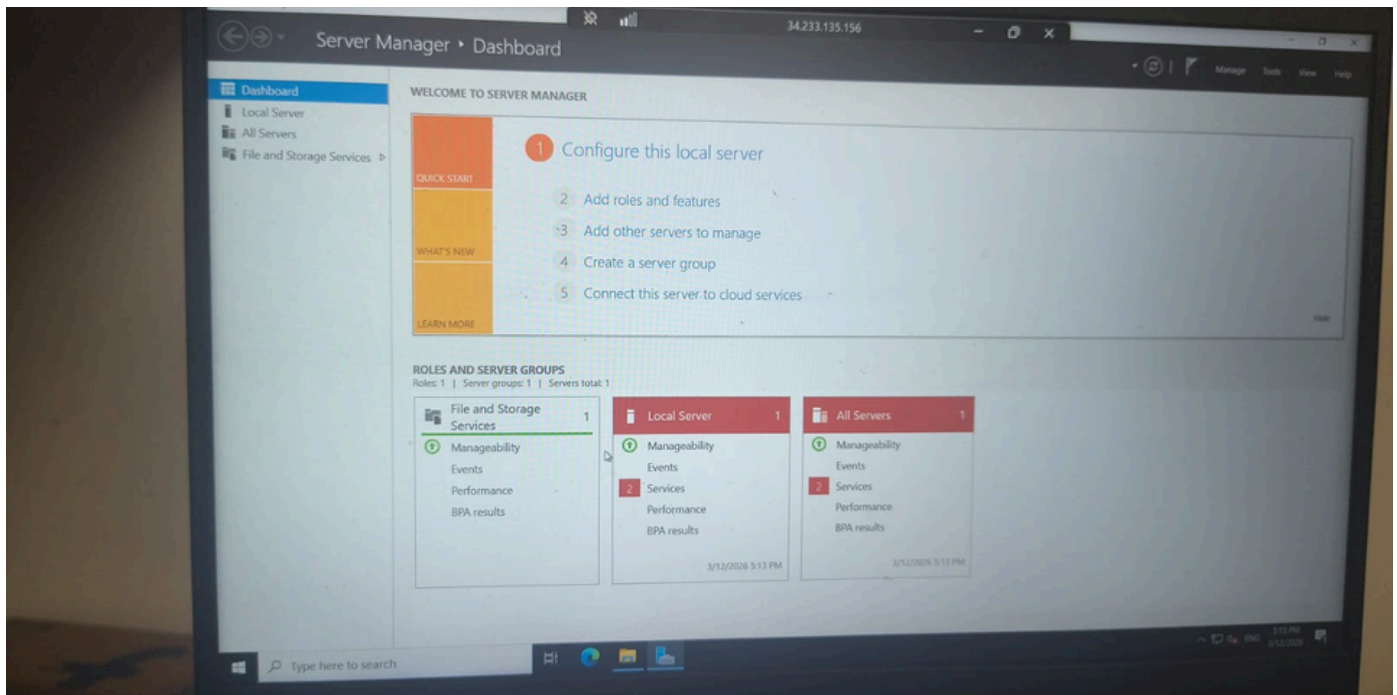
How to Configure the IIS web server in Windows Server with AWS EC2



Execution Tasks:



Execution Tasks:

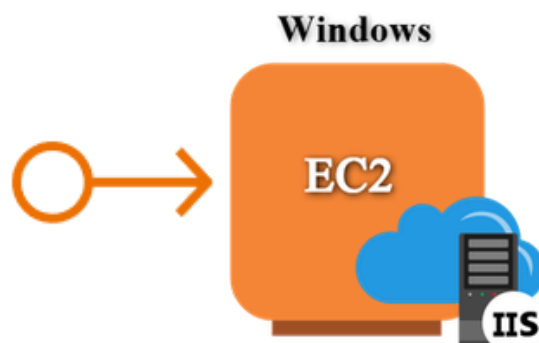




AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Capstone Projects

Trainer: Aravindraaj.G- Nminds Academy
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3.Configure Elastic IP Address to Windows WebServer in AWS



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Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

1. Highly Available Applications:

- o If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.

2. Web Servers:

- o If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.

3. Disaster Recovery:

- o Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

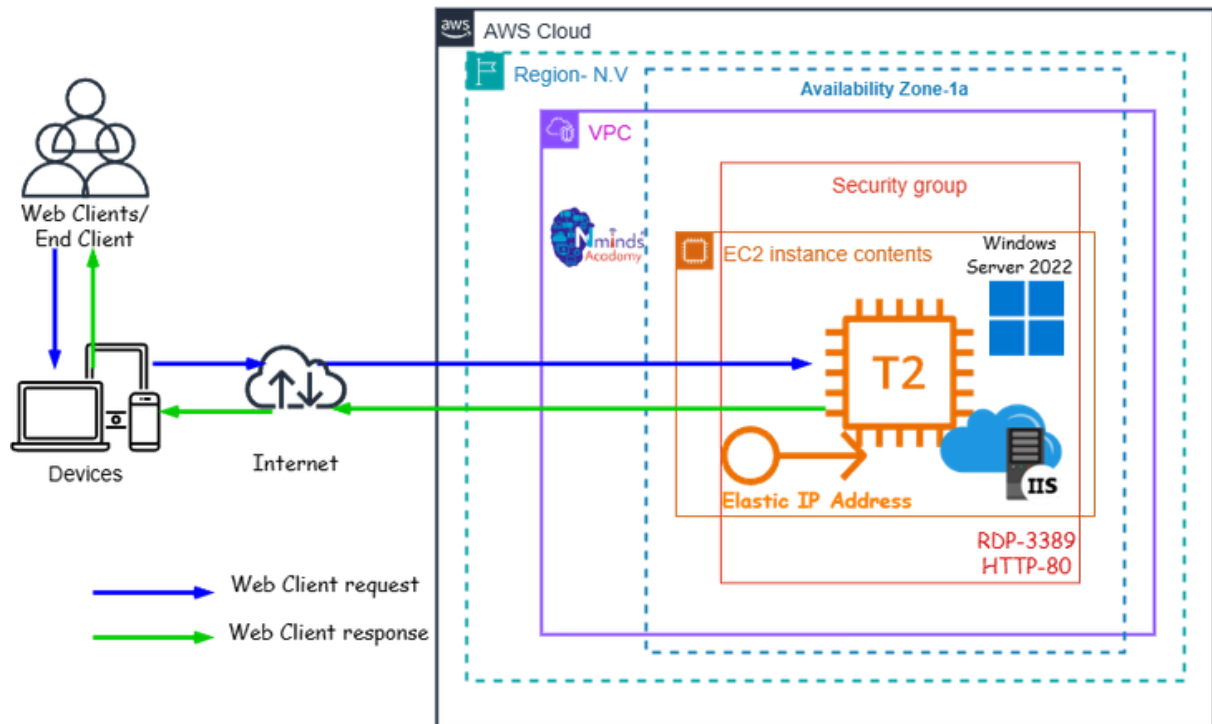
Best Practices:

- o Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- o Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- o Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.



Topology

How to Configure the Elastic IP Address to Windows Web Server with AWS EC2



Used Services in AWS

Used Services in Windows Server



Amazon EC2



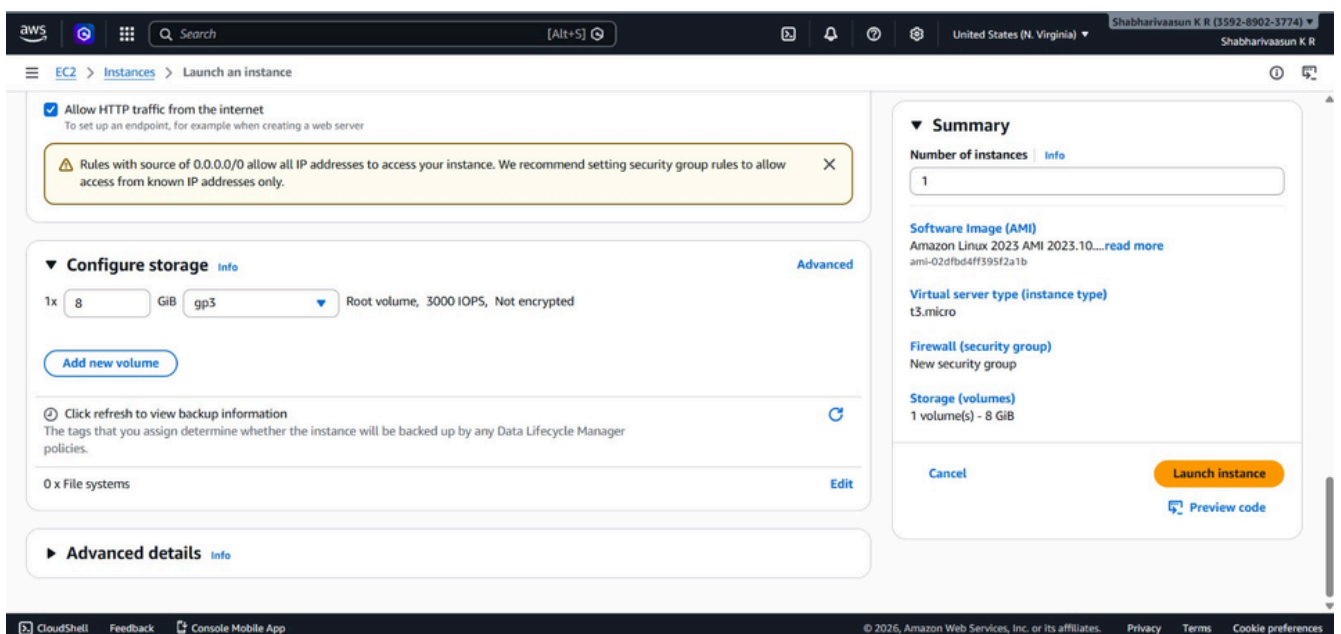
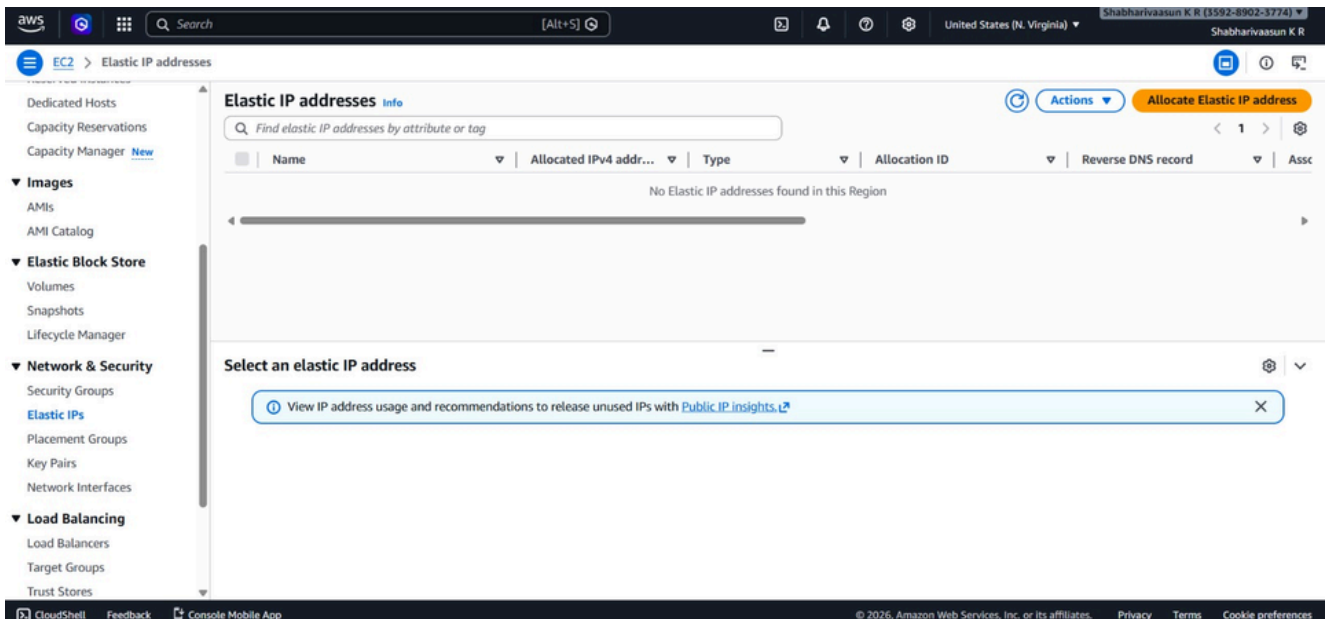
IIS- Internet Information Services



Amazon Elastic IP Address



Execution Tasks:



Execution Tasks:

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and the user's name 'Sowmighakala'. The left sidebar contains various service categories like EC2, Elastic Block Store, and Network & Security. The main content area is titled 'Elastic IP addresses' and shows a success message: 'Elastic IP address associated successfully. Elastic IP address 3.234.8.106 has been associated with instance i-05bad613f6a3d9302'. Below this, there's a table of Elastic IP addresses with columns for Name, Allocated IPv4 address, Type, Allocation ID, Reverse DNS record, and Associated Instance ID. The table shows one entry for the IP 3.234.8.106, which is a Public IP with Allocation ID eipalloc-0290ff0b8d3d6ac82, associated with instance i-05bad613f6a3d9302. Below the table, there's a detailed view for the selected IP address, showing its summary, type (Public IP), scope (VPC), and associated instance ID.

The screenshot shows a web browser window displaying a personal profile page titled 'About Me'. The page has a blue header and a white content area. The profile information is as follows:

- Hello!**
My name is Sowmigha. I am a Computer Science student interested in programming, machine learning, and software development. I enjoy building projects that solve real-world problems and improve people's lives.
- Education**
Currently pursuing B.E in Computer Science and Engineering (AI & Data Science). I am learning programming languages like C, C++, Java, and Python along with web technologies.
- Skills**
C, C++, Java, Python, HTML, CSS, Machine Learning
- Projects**
 - Weather Prediction Application
 - Smart Home Energy Usage Tracker
 - Campus Placement Prediction using Machine Learning
- Contact**
Email: yourmail@example.com
Location: Tamil Nadu, India





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4.Create an EC2 Linux Instance in AWS



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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache, FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

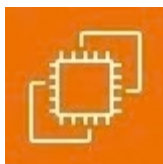
Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.

Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.

Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.

Development and Testing: Create development environments with linux OS to test php, MySQL or other Linux-based applications.

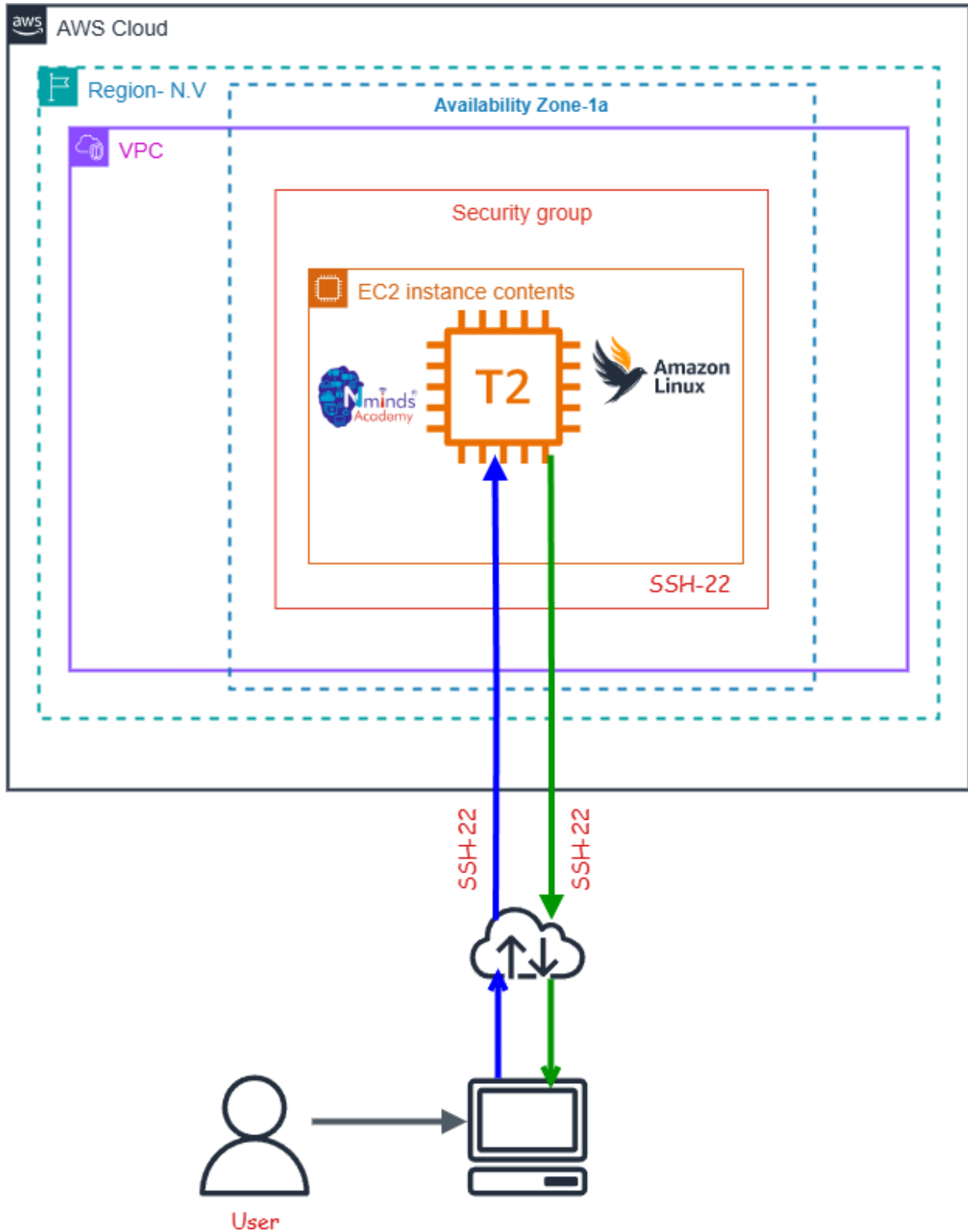
Used Services in AWS



Amazon EC2



Topology



Execution Tasks:

The screenshot shows the 'Launch an instance' page in the AWS Management Console. The top navigation bar includes the AWS logo, a search bar, and the region 'United States (N. Virginia)'. The left sidebar shows the 'EC2' menu with 'Instances' selected. The main content area is titled 'Launch an instance' and features a search bar for AMIs. Below this, there's a 'Quick Start' section with icons for various operating systems: Amazon Linux, macOS, Ubuntu, Windows (selected), Red Hat, SUSE Linux, and Debian. The 'Amazon Machine Image (AMI)' section displays the 'Microsoft Windows Server 2022 Base' AMI (ami-0f7a0c94dce9ab456) with details like 'Free tier eligible', 'Virtualization: hvm', 'ENA enabled: true', and 'Root device type: ebs'. A 'Description' section notes it's the 'Microsoft Windows 2022 Datacenter edition'. A table lists the 'Architecture' (64-bit (x86)), 'AMI ID', 'Publish Date' (2026-02-12), and 'Username' (Administrator). The right sidebar contains a 'Summary' section with fields for 'Number of instances' (set to 1), 'Software Image (AMI)', 'Virtual server type (instance type)' (t3.micro), 'Firewall (security group)' (New security group), and 'Storage (volumes)' (1 volume(s) - 30 GiB). At the bottom right, there are 'Cancel' and 'Launch instance' buttons, along with a 'Preview code' link.

The screenshot shows the 'Instances' page in the AWS Management Console. The top navigation bar is consistent with the previous image. The left sidebar shows the 'EC2' menu with 'Instances' selected. The main content area is titled 'Instances' and includes a search bar for finding instances by attribute or tag. Below the search bar, there's a table with columns: Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4. The table is currently empty, displaying the message 'No instances' and 'You do not have any instances in this region'. A 'Launch instances' button is visible. The bottom of the page shows the footer with '© 2026, Amazon Web Services, Inc. or its affiliates.' and links for 'Privacy', 'Terms', and 'Cookie preferences'.



Execution Tasks:

EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0cb52d1ebb0fad104)

Launch log

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts
Manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
[Create billing alerts](#)

Connect to your instance
Once your instance is running, log into it from your local computer.
[Connect to instance](#)
[Learn more](#)

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
[Connect an RDS database](#)
[Create a new RDS database](#)
[Learn more](#)

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots.
[Create EBS snapshot policy](#)

Manage detailed monitoring
[AddShell](#) [Feedback](#) [Console](#) [Mobile App](#)

Create Load Balancer **Create AWS budget** **Manage CloudWatch alarms**

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EC2 > Instances

Instances (1/2) Info
Last updated less than a minute ago
[Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 D
Project 1	i-0cb52d1ebb0fad104	Running	t3.micro	3/3 checks passed	...	us-east-1c	ec2-54-242-98
Project 2	i-0813ccc1fc44a9dd0	Running	t3.micro	...	...	us-east-1c	ec2-54-226-62

i-0cb52d1ebb0fad104 (Project 1)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-0cb52d1ebb0fad104	Public IPv4 address 54.242.98.204 open address	Private IPv4 addresses 172.31.22.200
IPv6 address -	Instance state Running	Public DNS ec2-54-242-98-204.compute-1.amazonaws.com open address
Instance type t3.micro	Private IP DNS name (IPv4 only)	





AWS Solution Architect Training with
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AWS Academy Certified Educator

5.Configure Apache Web Server on Linux Instance in X AWS



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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache, FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.

Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.

Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.

Development and Testing: Create development environments with linux OS to test php, MySql or other Linux-based applications.

Used Services in AWS



Amazon EC2

Used Services in Linux Server

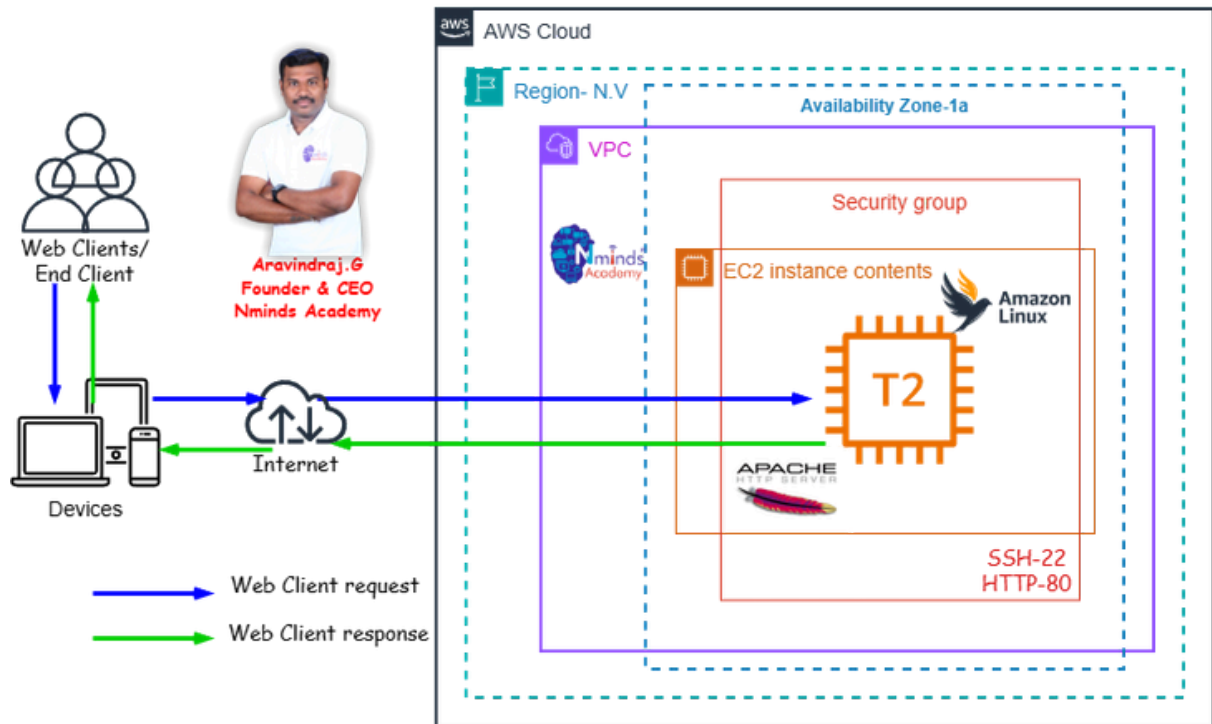


Apache Webserver



Topology

How to Configure the Apache web server in Linux Server with AWS EC2



Execution Tasks:

The screenshot displays the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the user's profile. The left sidebar shows the navigation menu with categories like EC2, IAM, and CloudFormation. The main content area is titled 'Instances (1/2) Info' and shows a table of EC2 instances. Two instances are listed: 'Project 1' (ID: i-0cb52d1ebb0fad104) and 'Project 2' (ID: i-0813ccc1fc44a9dd0). Both are in a 'Running' state. Below the table, the details for 'Project 1' are expanded, showing its instance ID, IP addresses, and status.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 D?
Project 1	i-0cb52d1ebb0fad104	Running	t3.micro	3/3 checks passed	...	us-east-1c	ec2-54-242-98-204
Project 2	i-0813ccc1fc44a9dd0	Running	t3.micro	...	...	us-east-1c	ec2-54-226-63-79

i-0cb52d1ebb0fad104 (Project 1)

Instance summary

Instance ID: i-0cb52d1ebb0fad104

IPv6 address: -

Public IPv4 address: 54.242.98.204 | [open address](#)

Instance state: Running

Private IPv4 addresses: 172.31.22.200

Public DNS: ec2-54-242-98-204.compute-1.amazonaws.com | [open address](#)

```
ec2-user@ip-172-31-16-241:~  
Microsoft Windows [Version 10.0.26200.8037]  
Microsoft Corporation. All rights reserved.  
  
C:\Users\shabh>cd downloads  
  
C:\Users\shabh\Downloads>ssh -i Shabhari.pem ec2-user@54.226.63.79  
Warning: Permanently added '54.226.63.79' (ED25519) to the list of known hosts.  
ec2-user@ip-172-31-16-241:~$
```



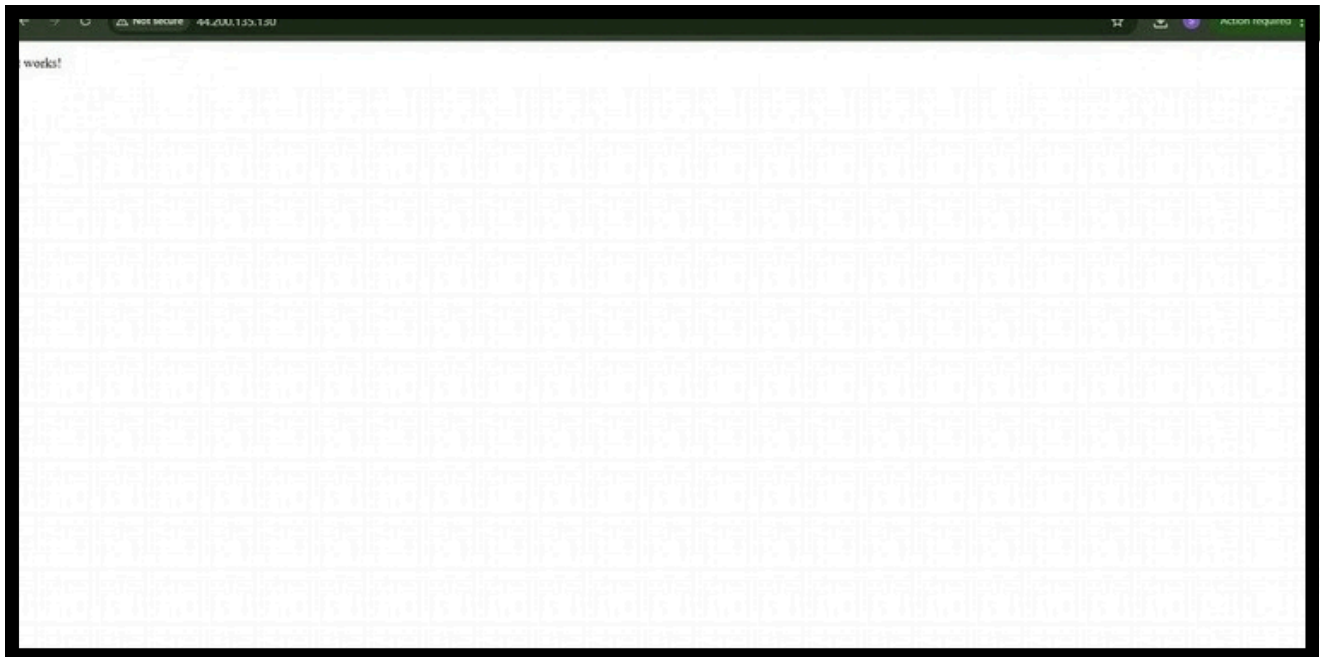
Execution Tasks:

```
root@ip-172-31-16-241:~  
Installing      : mod_http2-2.0.27-1.amzn2023.0.3.x86_64      10/13  
Installing      : mod_lua-2.4.66-1.amzn2023.0.1.x86_64      11/13  
Installing      : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13  
Installing      : httpd-2.4.66-1.amzn2023.0.1.x86_64        13/13  
Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64        13/13  
Verifying       : apr-1.7.5-1.amzn2023.0.4.x86_64           1/13  
Verifying       : apr-util-1.6.3-1.amzn2023.0.2.x86_64       2/13  
Verifying       : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64   3/13  
Verifying       : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13  
Verifying       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13  
Verifying       : httpd-2.4.66-1.amzn2023.0.1.x86_64        6/13  
Verifying       : httpd-core-2.4.66-1.amzn2023.0.1.x86_64    7/13  
Verifying       : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13  
Verifying       : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64   9/13  
Verifying       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64      10/13  
Verifying       : mailcap-2.1.49-3.amzn2023.0.3.noarch       11/13  
Verifying       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64     12/13  
Verifying       : mod_lua-2.4.66-1.amzn2023.0.1.x86_64      13/13  
  
Installed:  
apr-1.7.5-1.amzn2023.0.4.x86_64      apr-util-1.6.3-1.amzn2023.0.2.x86_64  
apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64  apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64  
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  httpd-2.4.66-1.amzn2023.0.1.x86_64  
httpd-core-2.4.66-1.amzn2023.0.1.x86_64  httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch  
httpd-tools-2.4.66-1.amzn2023.0.1.x86_64  libbrotli-1.0.9-4.amzn2023.0.2.x86_64  
mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.27-1.amzn2023.0.3.x86_64  
mod_lua-2.4.66-1.amzn2023.0.1.x86_64  
  
complete!  
root@ip-172-31-16-241 ~]# |
```

```
root@ip-172-31-72-105:~  
# Logging  
#SyslogFacility AUTH  
#LogLevel INFO  
  
# Authentication:  
  
#LoginGraceTime 2m  
#PermitRootLogin prohibit-password  
#StrictModes yes  
#MaxAuthTries 6  
#MaxSessions 10  
  
PubkeyAuthentication no  
  
# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2  
# but this is overridden so installations will only check .ssh/authorized_keys  
AuthorizedKeysFile      .ssh/authorized_keys  
  
#AuthorizedPrincipalsFile none  
  
# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts  
#HostbasedAuthentication no  
# Change to yes if you don't trust ~/.ssh/known_hosts for  
# HostbasedAuthentication  
#IgnoreUserKnownHosts no  
# Don't read the user's ~/.rhosts and ~/.shosts files  
#IgnoreRhosts yes  
  
# Explicitly disable PasswordAuthentication. By presetting it, we  
# avoid the cloud-init set_passwords module modifying sshd_config and  
# restarting sshd in the default instance launch configuration.  
PasswordAuthentication yes  
PermitEmptyPasswords no  
  
# Change to no to disable s/key passwords  
#KbdInteractiveAuthentication yes  
  
# Kerberos options  
#KerberosAuthentication no  
-- INSERT --  
  
65,27 34%
```



Execution Tasks:

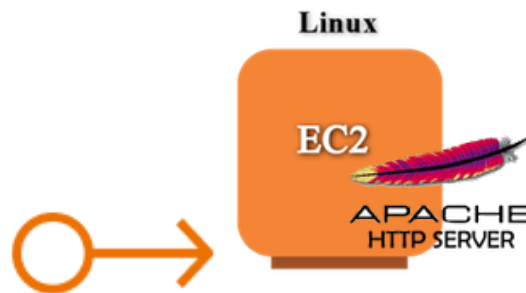




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AWS Academy Certified Educator

6.Configure Elastic IP Address to Windows Web Server in AWS



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Sri Eshwar College of Engineering,
Coimbatore.



Objective

An Elastic IP (EIP) in AWS is a static, public IP address designed for dynamic cloud computing. It is associated with your AWS account and can be quickly associated or disassociated with any EC2 instance in your account. This feature is particularly useful when you need to maintain a consistent IP address for your resources, even when you stop and start EC2 instances.

Common Use Cases for Elastic IP:

1. Highly Available Applications:

- o If you're running a service that requires high availability, you can use Elastic IPs to quickly reassign a static IP to a new instance if your primary instance fails, ensuring minimal downtime.

2. Web Servers:

- o If you host a website and need to ensure the IP address remains the same even if the underlying EC2 instance is restarted, an Elastic IP helps maintain this consistency.

3. Disaster Recovery:

- o Elastic IPs are useful for disaster recovery scenarios. If one instance goes down, you can quickly associate the EIP with a backup instance to ensure services are still accessible.

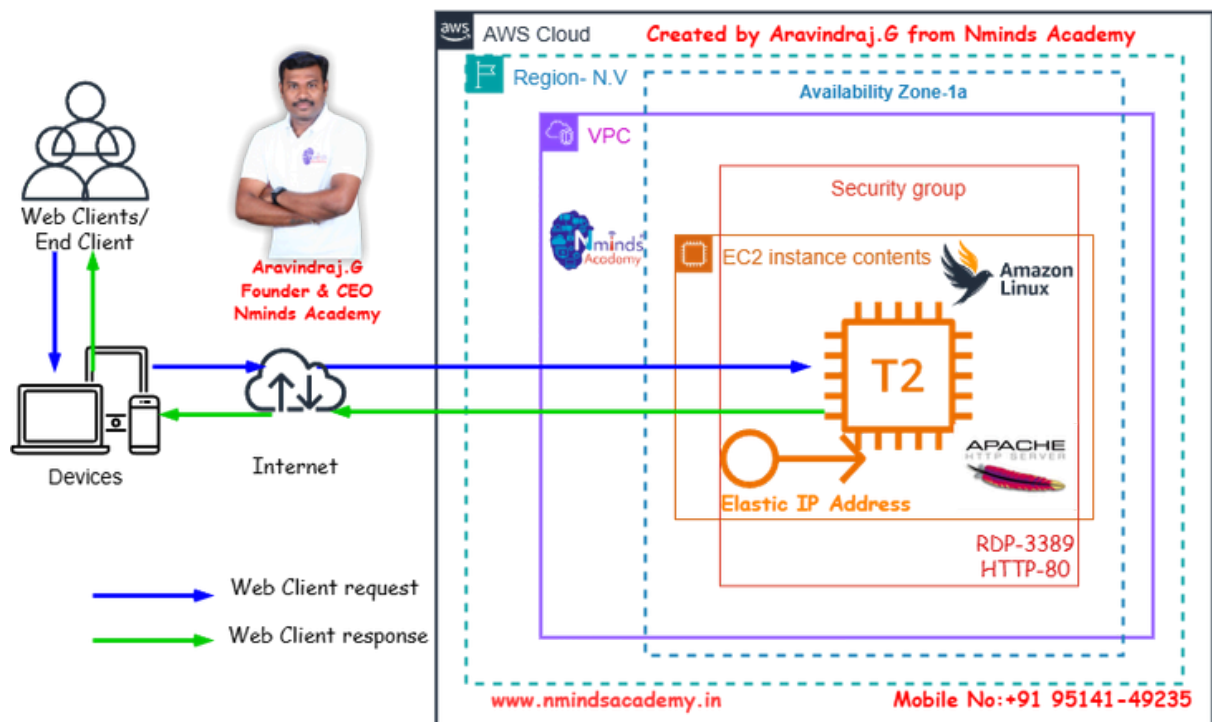
Best Practices:

- o Use Elastic IPs only when necessary: Since they come with associated costs when unused, it's a good practice to release EIPs that are no longer required.
- o Move IPs during instance failure: Instead of keeping an EIP permanently attached, use it as a failover method, reassociating it to a new instance when needed.
- o Monitor EIP Usage: Periodically review your usage to ensure you're not paying for unnecessary Elastic IP addresses.



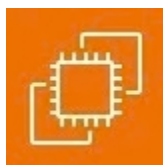
Topology

How to Configure the Elastic IP Address to Linux Web Server with AWS EC2



Used Services in AWS
Services in Windows Server

Used



Amazon EC2



Apache Webserver



Amazon ElasticIPAddress



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Execution Tasks:

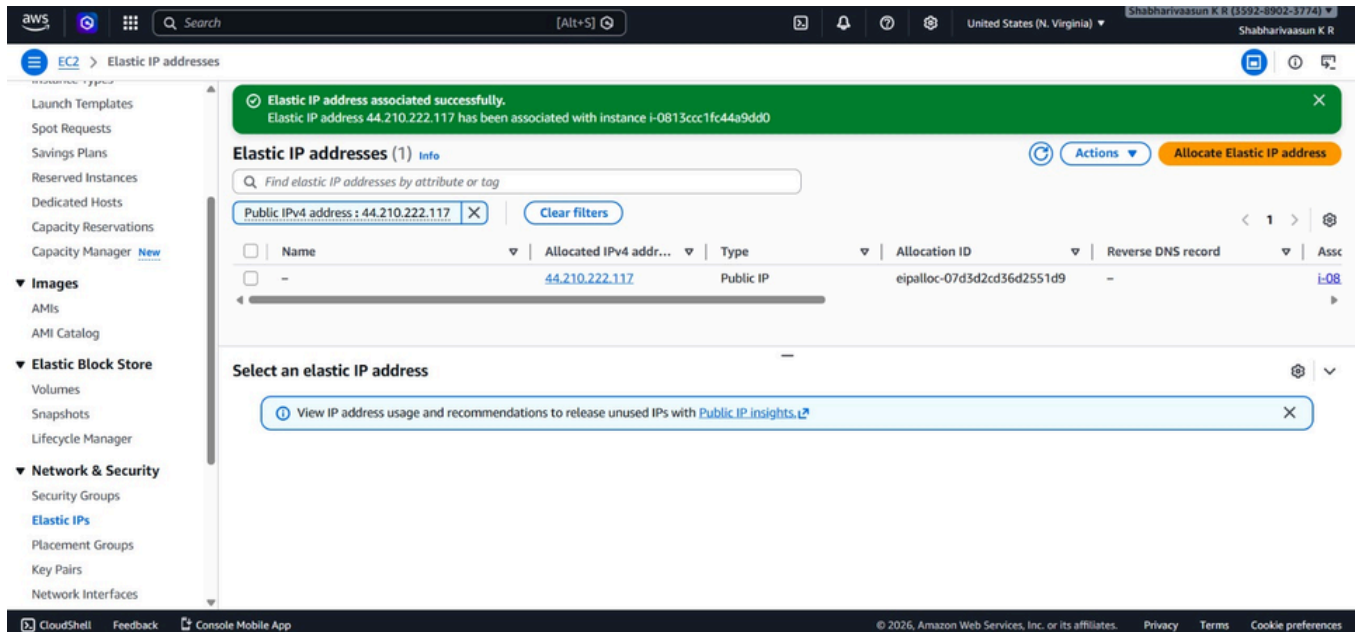
The screenshot shows the 'Allocate Elastic IP address' page in the AWS console. The breadcrumb navigation is 'EC2 > Elastic IP addresses > Allocate Elastic IP address'. The page title is 'Allocate Elastic IP address' with an 'info' link. Under 'Elastic IP address settings', the 'Public IPv4 address pool' is selected. The 'Network border group' is set to 'us-east-1'. A section for 'Global static IP addresses' mentions AWS Global Accelerator. A 'Tags - optional' section is also present. The footer includes 'CloudShell', 'Feedback', 'Console Mobile App', and copyright information for Amazon Web Services, Inc.

The screenshot shows the 'Elastic IP addresses' page in the AWS console. A green banner at the top states 'Elastic IP address allocated successfully. Elastic IP address 44.210.222.117'. The page title is 'Elastic IP addresses (1)' with an 'info' link. A search bar and 'Clear filters' button are visible. A table lists the allocated IP address. The footer includes a sidebar with navigation links like 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', 'Dedicated Hosts', 'Capacity Reservations', 'Capacity Manager', 'Images', 'Elastic Block Store', and 'Network & Security'.

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associate
-	44.210.222.117	Public IP	eipalloc-07d3d2cd36d2551d9	-	-



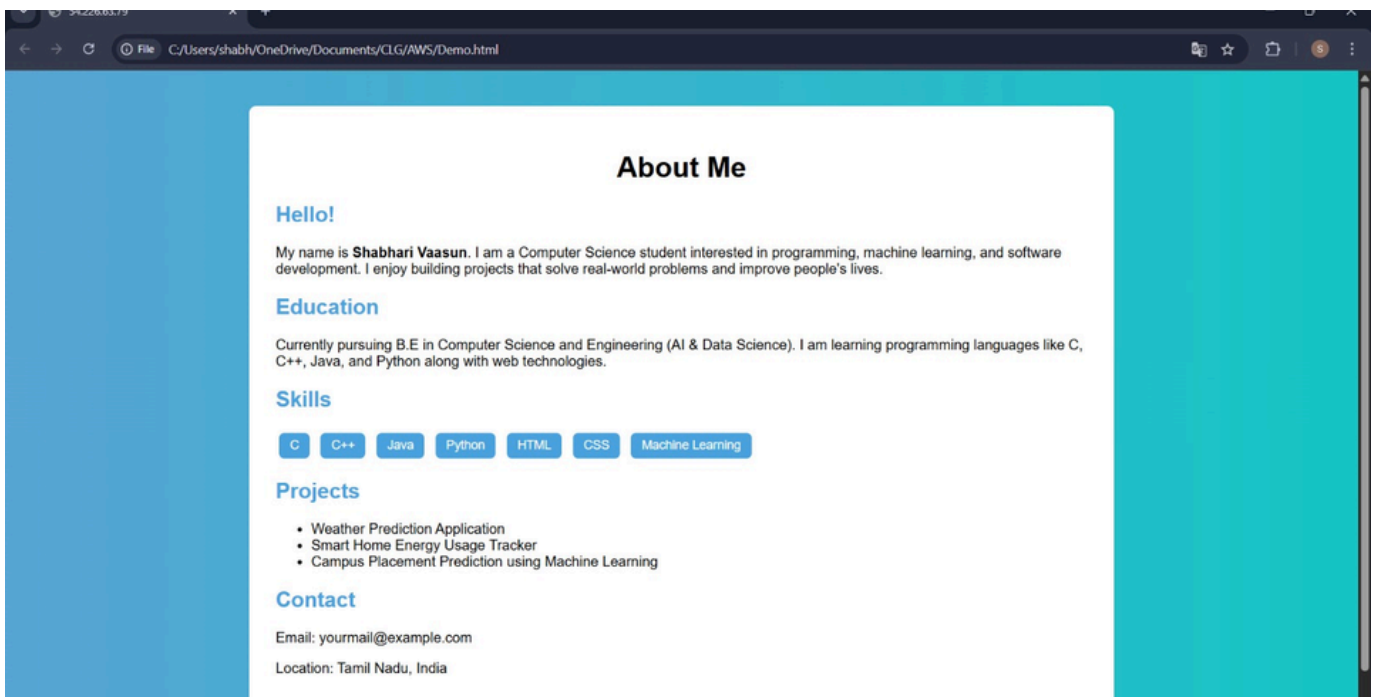
Execution Tasks:



The screenshot shows the AWS Management Console interface. At the top, a green notification banner states: "Elastic IP address associated successfully. Elastic IP address 44.210.222.117 has been associated with instance i-0813ccc1fc44a9dd0". Below this, the "Elastic IP addresses (1)" section is visible. It includes a search bar and a table with one entry:

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associated Instance
-	44.210.222.117	Public IP	eipalloc-07d3d2cd36d2551d9	-	i-0813ccc1fc44a9dd0

Below the table, there is a section "Select an elastic IP address" with a link to "View IP address usage and recommendations to release unused IPs with Public IP insights".



The screenshot shows a web browser displaying a personal profile page titled "About Me". The page has a teal background and white text. The content includes:

About Me

Hello!

My name is **Shabhari Vaasun**. I am a Computer Science student interested in programming, machine learning, and software development. I enjoy building projects that solve real-world problems and improve people's lives.

Education

Currently pursuing B.E in Computer Science and Engineering (AI & Data Science). I am learning programming languages like C, C++, Java, and Python along with web technologies.

Skills

C C++ Java Python HTML CSS Machine Learning

Projects

- Weather Prediction Application
- Smart Home Energy Usage Tracker
- Campus Placement Prediction using Machine Learning

Contact

Email: yourmail@example.com
Location: Tamil Nadu, India





AWS Solution Architect Training with AWS Cloud Practitioner Global Certification Capstone Projects

Trainer: Aravindraaj.G- Nminds Academy
AWS Academy Certified Educator

7.Configure SFTP toLinux Web Server in AWS



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CSE-Artificial Intelligence & Machine Learning,
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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

Use Cases for EC2 Linux Instances:

Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.

Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.

Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.

Development and Testing: Create development environments with linux OS to test php, MySQL or other Linux-based applications.

SFTP- Secure File Transfer Protocol

SFTP (Secure File Transfer Protocol) is a secure network protocol that allows users to transfer files over a secure, encrypted connection. It is a more secure alternative to FTP (File Transfer Protocol) and is commonly used to upload, download, and manage files on remote servers.

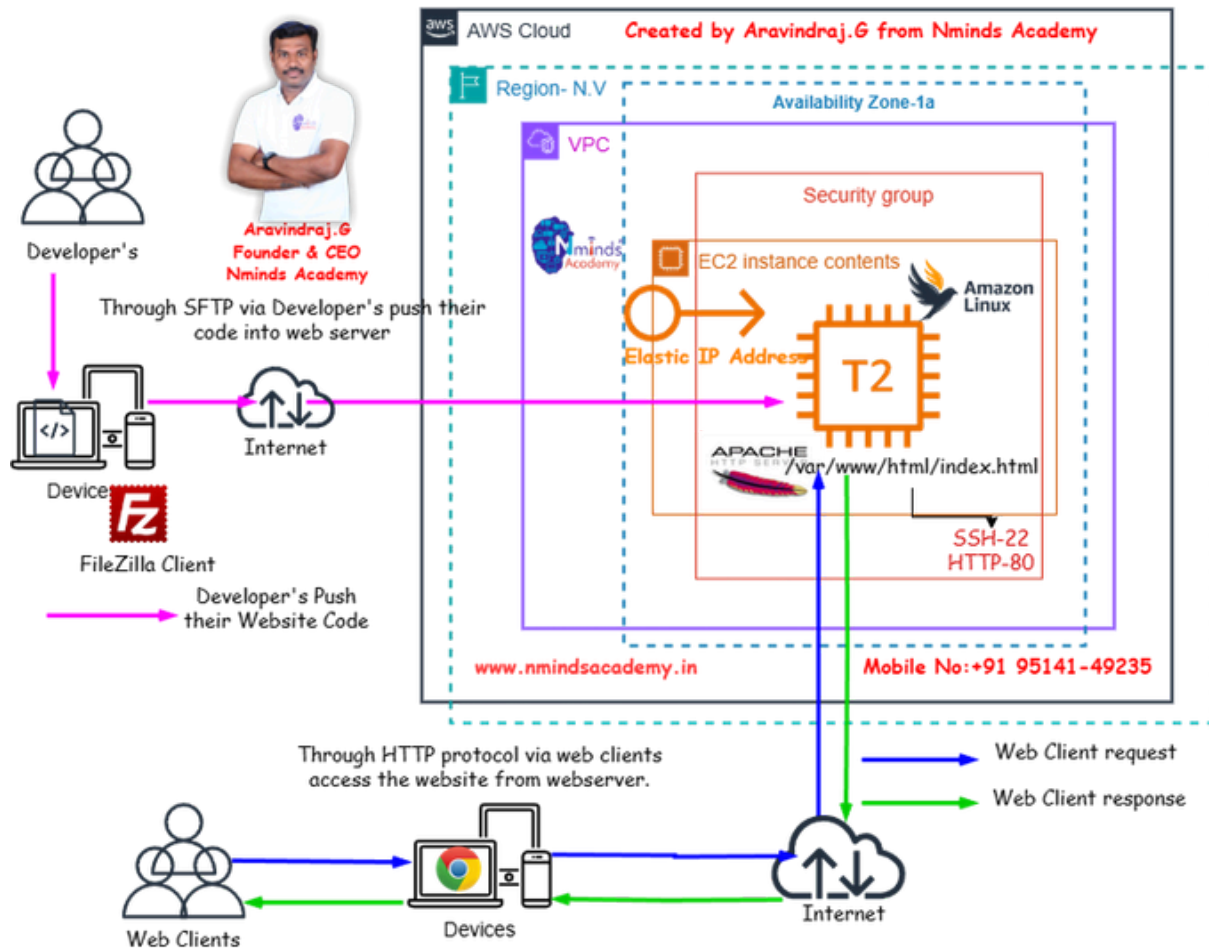
Here are the key uses and advantages of SFTP:

1. **File Transfer:** SFTP allows secure transfer of files between a local and a remote computer over a network. It ensures that the data is encrypted during transmission, protecting it from eavesdropping or tampering.
2. **File Management:** Besides transferring files, SFTP also allows users to perform file management tasks such as renaming, deleting, and moving files on the remote server.
3. **Secure Authentication:** SFTP uses SSH (Secure Shell) for authentication, which provides a stronger level of security compared to FTP's traditional username/password system. It often supports public key authentication, adding an extra layer of security.
4. **Encryption:** The main advantage of SFTP over FTP is encryption. All data sent over SFTP is encrypted, ensuring that sensitive information (like passwords and files) is protected during transmission.
5. **Firewall-Friendly:** SFTP operates over a single port (usually port 22), making it easier to use behind firewalls and NAT (Network Address Translation) devices compared to FTP, which uses multiple ports.



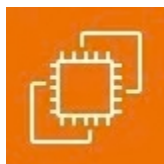
Topology

How to Configure Developer's push their website code by using SFTP Protocol in Linux Server with AWS EC2 with Elastic IP Address



Used Services in AWS
Services in Windows Server

Used



Amazon EC2



Apache Webserver



Amazon Elastic IP Address



Execution Tasks:

```

ec2-user@ip-172-31-16-241:~ + ~
w] 50L, 12918 written
t@ip-172-31-16-241 html]# client_loop: send disconnect: Connection reset

sers\shabh\Downloads>ssh -i Shabahri.pen ec2-user@44.210.222.117
ing: Identity file Shabahri.pen not accessible: No such file or directory.
authenticity of host '44.210.222.117 (44.210.222.117)' can't be established.
519 key fingerprint is SHA256:SUKMwbNsfnKwE5ejPEpDQxVmndHtg316T3QoZs1o/Y.
host key is known by the following other names/addresses:
C:\Users\shabh/.ssh/known_hosts:1: 54.226.63.79
you sure you want to continue connecting (yes/no/[fingerprint])? yes
ing: Permanently added '44.210.222.117' (ED25519) to the list of known hosts.
user@44.210.222.117: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).

sers\shabh\Downloads>ssh -i Shabhari.pen ec2-user@44.210.222.117
ing: Identity file Shabhari.pen not accessible: No such file or directory.
user@44.210.222.117: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).

sers\shabh\Downloads>ssh -i Shabhari.pem ec2-user@44.210.222.117
#_
\_ #####_ Amazon Linux 2023
\_#####\
\_###|
\_#/ --- https://aws.amazon.com/linux/amazon-linux-2023
V~' ' ->
login: Thu Mar 12 18:05:49 2026 from 171.79.58.49
-user@ip-172-31-16-241 ~]$ |

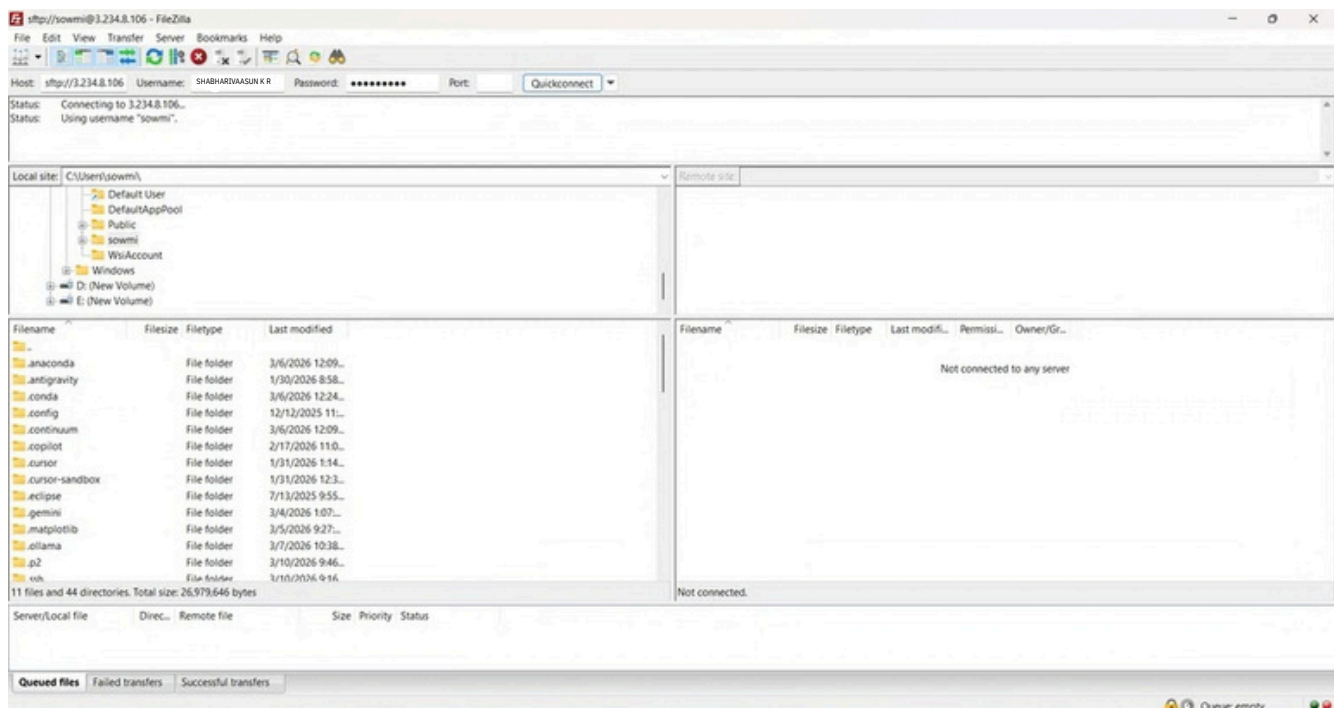
```

```
root@ip-172-31-16-241:~  
daemon:x:2:2:daemon:/sbin:/sbin/nologin  
adm:x:3:4:adm:/var/adm:/sbin/nologin  
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin  
sync:x:5:0:sync:/sbin:/bin/sync  
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown  
halt:x:7:0:halt:/sbin:/sbin/halt  
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin  
operator:x:11:0:operator:/root:/sbin/nologin  
games:x:12:100:games:/usr/games:/sbin/nologin  
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin  
nobody:x:65534:65534:Kernel Overflow User:/sbin/nologin  
dbus:x:81:81:System message bus:/sbin/nologin  
systemd-network:x:192:192:systemd Network Management:/usr/sbin/nologin  
systemd-oom:x:998:998:systemd Userspace OOM Killer:/usr/sbin/nologin  
systemd-resolve:x:193:193:systemd Resolver:/usr/sbin/nologin  
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin  
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/usr/sbin/nologin  
systemd-coredump:x:996:996:systemd Core Dumper:/usr/sbin/nologin  
systemd-timesync:x:995:995:systemd Time Synchronization:/usr/sbin/nologin  
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin  
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin  
ec2-instance-connect:x:993:993:./home/ec2-instance-connect:/sbin/nologin  
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin  
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin  
tcpdump:x:72:72:./:/sbin/nologin  
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash  
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin  
dev1deeps:x:1001:1001:./home/dev1deeps:/bin/bash  
dev2deeps:x:1002:1002:./home/dev2deeps:/bin/bash  
root@ip-172-31-16-241 ~]#
```



Execution Tasks:

```
root@ip-172-31-72-105:~  
# Logging  
#SyslogFacility AUTH  
#LogLevel INFO  
  
# Authentication:  
#LoginGraceTime 2m  
#PermitRootLogin prohibit-password  
#StrictModes yes  
#MaxAuthTries 6  
#MaxSessions 10  
  
PubkeyAuthentication no  
  
# The default is to check both .ssh/authorized_keys and .ssh/authorized_keys2  
# but this is overridden so installations will only check .ssh/authorized_keys  
AuthorizedKeysFile .ssh/authorized_keys  
  
#AuthorizedPrincipalsFile none  
  
# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts  
#HostbasedAuthentication no  
# Change to yes if you don't trust ~/.ssh/known_hosts for  
# HostbasedAuthentication  
#IgnoreUserKnownHosts no  
# Don't read the user's ~/.rhosts and ~/.shosts files  
#IgnoreRhosts yes  
  
# Explicitly disable PasswordAuthentication. By presetting it, we  
# avoid the cloud-init set_passwords module modifying sshd_config and  
# restarting sshd in the default instance launch configuration.  
PasswordAuthentication yes  
PermitEmptyPasswords no  
  
# Change to no to disable s/key passwords  
#KbdInteractiveAuthentication yes  
  
# Kerberos options  
#KerberosAuthentication no  
-- INSERT --
```



Execution Tasks:

